

UZAIR AKBAR

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SUMMARY

Software Engineer with experience in data eng., data science, and DevOps, focused on model deployment, scalable ETL pipelines, and CI/CD. Recent M.Sc. Computer Science graduate from Georgia Tech, specializing in Robust and Causal ML[†], with a strong interest in data/ML eng., cloud-based microservices, and Kubernetes-integrated systems.

EDUCATION

M.Sc. Computer Science @ Georgia Institute of Technology **GPA: 3.78 / 4.0 | Dec. 2025**
• **Teaching Assistant:** Reinforcement Learning ([CS8803](#), [CS7642](#) Fall '24 — present).
• **Coursework:** Machine Learning, Systems for ML, Database Systems, Computer Networks, OS, Algorithms
M.Sc. Communications Eng. @ Technical University of Munich, Germany **GPA: 3.65 / 4.0 | June 2021**
B.Sc. Electrical Eng. @ National University of Sciences & Technology, Pakistan **GPA: 3.59 / 4.0 | June 2017**

SELECTED PUBLICATIONS

[†]**U. Akbar et al.** “Symmetry-Informed Causal Partial Identification.” *under review @ ICLR*, 2027. [📄](#) [🌐](#)
[†]**U. Akbar et al.** “An Analysis of Causal Effect Estimation using Outcome Invariant Data Augmentation.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. (**Spotlight; top 3%**) [📄](#) [🌐](#) [🌐](#)
A. Savkin, T. Lapotre, K. Strauss, **U. Akbar** & F. Tombari, “Adversarial Appearance Learning in Augmented Cityscapes for Pedestrian Recognition in Autonomous Driving.” *ICRA*, 2020. [🌐](#)

PROFESSIONAL EXPERIENCE

Teaching Assistant (Reinforcement Learning) @ *Georgia Tech, USA* [📄](#) [🌐](#) **Aug. 2024 — Present**
• Authored RL projects for ~ 300 students/semester, including a `python` package for local [DeepRacer](#) deployment:
• Built a containerized simulation service with multi-runtime support (`docker` (`compose`) + `podman` + `aptainer`).
• Developed a client-side `gymnasium` wrapper to interface with the simulation service using `ZeroMQ` sockets.

Software Engineer Intern @ *Hewlett Packard Enterprise, USA* **June 2024 — Aug. 2024**
• Built CI/CD pipeline (`python` + `jenkins` + `JFrog`) to test/build/release/rollback VM templates for QA workloads.
• Integrated pipeline with dynamic KVM provisioning platform (`k8s` + `KubeVirt`) for scalable env. management.

Software Engineer @ *OMMAX, Germany* **Nov. 2022 — June 2023**
• Designed end-to-end Salesforce ETL pipeline (`Airflow` + `PostgreSQL`), improving data throughput by 58% (by refactoring `python` → `postgres`) and sales lead identification by 63% (by reducing data redundancy & duplicates).
• Built a serverless `FastAPI` microservice on AWS Lambda (`docker`) for `XGBoost`-based ticket classification and RegEx error code extraction. Served endpoints to a CRM platform to automate ticket routing and categorization.

Data Scientist @ *Auto1 Group, Germany* **June 2022 — Oct. 2022**
• Engineered `LightGBM` predictor for car refurbishment costs; €160 MAE decrease (~27.5% improvement) vs. prod.
• Automated training and eval. workflows; data extraction/transformation (Redshift) & model checkpointing (S3).

ML Research Intern @ *Max Planck Institute of Intelligent Systems, Germany* **Apr. 2020 — Feb. 2021**
• Developed a method to reduce hidden confounding bias w/ data-augmentation + instrumental variable regression.
• Basis of my Master's [thesis](#) @ TUM, [spotlight](#) @ NeurIPS 2025 and ongoing work in *geometric causal inference*.

TECH. STACK

Languages: `Python`, `C++`, `SQL`, `Bash` **Data & Backend:** `PostgreSQL`, `Airflow`, `Redis`, `FastAPI`
Testing & CI/CD: `Jenkins`, `JFrog`, `Git` **Systems & Orchestration:** `Docker`, `Linux`, `RabbitMQ`, `K8s`
Cloud: `AWS` (EC2, EKS, Lambda, S3, RDS, Redshift)